

SEDAR 65 HMS Atlantic Blacktip Shark

Data Workshop

Indices Work group

1. Explore the utility of combining multiple indices into one index using the Bayesian hierarchical model (Conn, 2009) or other similar methodology. The data series that could potentially be combined are:
 - For Age 0
 - Coastspan Longline, Coastspan Gillnet Short Net, Coastspan Gillnet Long Net
 - For All Ages
 - NEFSC Bottom Longline, Shark Bottom Longline Observer, Virginia Institute of Marine Science, SEAMAP Longline, SCDNR Red Drum Longline
2. Investigate alternate methods in future assessments for standardizing indices of abundances outside the Delta-Lognormal method (Lo et al. 1992).
3. Explore the utility of standardized age-0 indices as recruitment indices in the stock assessment model.

Life History Workgroup

- 1 Increase sampling intensity throughout range, particularly at depths less than 20 m.
- 2 Investigate sex- and life stage-specific movements of blacktip sharks to determine if migratory behaviors change based on maturity or reproductive condition.
- 3 Animals should be tagged throughout their range, including the northern extent of the population range off New York, to gain a more complete understanding of migratory and residency patterns.
- 4 Identify environmental conditions (e.g. dissolved oxygen, temperature, salinity, etc.) and ecological factors (e.g. prey abundance, community structure, etc.) that correlate with migration, movement patterns, and preferred habitats. This will allow prediction of future range changes based on habitat suitability models.
- 5 Identification of population structure based on genetic information or other intrinsic natural markers/tracers.

Catches Workgroup

1. Increase public education outreach activities for species identification in the recreational fishery. This is important because the fishery has become largely recreational, there are no species identification training workshops for recreational fishers, and it is difficult to distinguish blacktip from spinner sharks, especially as juveniles, by non-trained individuals.
2. Improve the MRIP process to filter biased sampling that leads to unreal, extreme fluctuations in catch data for sharks, through a QA step that is applied with an objective, non-arbitrary procedure.

Ecological Factors Working group

1. Quantify seasonal and spatial distribution of prey for Atlantic blacktip sharks, and use stomach contents analysis to determine the relative importance of different forage fish species in the diet. This is important in the New York Bight area where blacktip sharks were not previously abundant and are now exploiting resources that have not been previously subjected to this level of exploitation. It might also be important in the southern end of their range because, although anglers state that blacktip sharks are following baitfish down the coast, the peak in baitfish abundance occurs a few months before the blacktip sharks arrive off south Florida.
2. Model the effects of changing stock distribution, due to ecological factors, on the results of fixed-station, fisheries-independent surveys for stock assessment. In general such surveys assume that changes in relative abundance are a result of changing stock size, rather than shifts in range and distribution as a result of ecological change. Modeling how ecological factors affect stock distribution allows for better quantification of stock abundance as measured by fixed-station surveys.
3. Conduct research on ecological changes in blacktip shark inshore nursery areas on the U.S. Atlantic coast and how those changes have affected recruitment.
4. Assess the levels of environmental contaminants in blacktip sharks and how those affect the sharks' physiology and reproductive success.
5. Study the response of blacktip sharks to harmful algal blooms and how those phenomena affect the status of the Atlantic stock of these sharks.

Assessment Process

Additional research may be needed on the variable effects of Federal and state recreational management actions on the annual length composition of Atlantic blacktip shark recreational catch ($A + B1 + B2\text{-Dead}$). During Assessment Webinars I and III, it was discussed that data limitations resulting from recreational length sampling might not accurately reflect the effect of Federal management actions on length composition of retained and discarded Atlantic blacktip sharks over time. Federal management actions include implementation of a minimum size limit (54 inches straight fork length) in Federal waters during calendar years 2000 – 2018 and the implementation of Federal bag limits of 4 LCS (Large Coastal Sharks; 1993), 2 LCS (1997) and 1 LCS (2000 – 2018). It was also noted that most Atlantic blacktip sharks are captured recreationally within state waters, and that the Federal management actions identified above may not have been implemented uniformly within state waters.

The selectivity parameterization approach implemented here estimated selectivity parameters where possible and fixed (or reformulated) poorly estimated selectivity parameters where necessary. This pragmatic selectivity parameterization approach is consistent with regularization to reduce over-parameterization in Bayesian stock assessments implemented in AD Model Builder, ADMB, by adding priors and turning off estimation for poorly informed parameters (Monnahan et al. 2019). This pragmatic approach was implemented here in order to remove sharks from the modeled population at the correct aggregate size sampled by each data set (**Figures 3.4 and 3.B.3**), while allowing relatively poorer fits to some poor quality annual length composition data sets (e.g., because of low sample size; **Figures 3.3 and 3.B.2**). An assumption was that poor quality annual length composition data sets were not necessarily representative of annual changes in the length composition sampled in that year (e.g., because of low sample size and observation error). In contrast, the aggregate length composition data obtained from the poor quality data were assumed to be representative of the length composition

sampled in that data set (e.g., because of higher sample size and reduced observation error in aggregate). Future research could investigate trade-offs in model fit and uncertainty by evaluating selectivity functions with fewer parameters and developing informed priors for the selectivity parameters.

The observation of proportionally few large sharks in the sampled length composition data compared to that expected based on life history may result for reasons other than dome-shaped selectivity. For example, the spatial distribution of fishing effort for an exploited population that is not well mixed (Sampson 2014) and selection of individuals with relatively faster growth rates (Taylor and Methot 2013) can also produce apparent dome-shaped selectivity patterns if not explicitly accounted for. Alternative modelling approaches for dealing with apparent dome-shaped selectivity can result in different underlying population numbers at age predicted over time within the stock assessment model. An attempt was made here to evaluate the effect of uncertainty in selectivity for fleets F1 and F2 and surveys S4 and S7 on the underlying population numbers at age predicted over time within the stock assessment model by implementing logistic selectivity for F1, F2, S4, and S7 within the logistic sensitivity analysis.

A growing number of model diagnostic methods are becoming available for use in integrated stock assessment models such as Stock Synthesis (e.g., Maunder and Piner 2015, 2017; Carvalho et al. 2017). Examples of implementing some of these diagnostic methods were provided as reference document (SEDAR65-RD13; Courtney et al. 2020). However, this set of diagnostics was not implemented within the current assessment due to time constraints. Additional research is also ongoing to improve the interpretation of model diagnostics in both model development and in model selection for use in providing management advice. For example, Maunder et al (2020) describe a risk-based approach based on individual model diagnostic results that assigned different weights to models used for management advice within an ensemble of candidate models.

Reproductive output timing within the Stock Synthesis assessment model is an active area of investigation within the SEFSC PCL stock assessment enterprise. In older versions of Stock Synthesis (< v3.30), implemented for Atlantic HMS SEDAR shark stock assessments, spawning stock size was calculated annually at the beginning of one specified spawning season and this spawning stock size produced one annual total recruitment value. Our intent in Stock Synthesis version 3.30 had been to change both the spawning timing (to June) and recruitment timing (to July). However, preliminary model runs with spawning timing defined as June (month 6) and recruitment timing defined as July (month 7) crashed, and require further evaluation before this setup can be implemented. In addition, recruitment is assumed to occur at age-0 in Stock Synthesis, consistent with previous Atlantic HMS SEDAR domestic shark stock assessments conducted with Stock Synthesis (Anon. 2015, 2017a, 2018). In contrast, recruitment was assumed to occur at age-1 in Atlantic HMS SEDAR domestic shark stock assessments previously conducted with a SSASPM (Anon. 2012, 2013a, 2013b).

Model sensitivity to reproductive output timing could be investigated in the future assessments. For example, defining the real age associated with L_{Amin} as age-1 and the size at the parameter value for L_{Amin} based on the VBG length at age-1 might be more consistent with previous SSASPM

implementations. However, in the length-based Stock Synthesis model implemented here, the recruitment timing and the resulting body size at recruitment also interact with other parameters within the Stock Synthesis model such as the CV in L_{Amin} , as well as with natural mortality and fishing mortality, which occur annually within the calendar year of recruitment. Consequently, an attempt was made here to evaluate model sensitivity to the combined effect of these interactions by estimating the CV in L_{Amin} within the logistic model sensitivity analysis described above.

Review Workshop

1. TOR 6 Consider the research recommendations provided by the Data Workshop and Assessment Process and make any additional recommendations or prioritizations warranted.

a. Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments.

The assessment team did explore the different ways of combining indices, as recommended from the data workshop. For age-0 the hierarchical Bayesian and dynamic factor analysis produced similar indices, so the latter was used. The inclusion in the assessment resulted in poor fit, non-convergence, or convergence to unreasonable parameter values. A subset of indices was used in a sensitivity analysis. The review panel shares the assessment panel's conclusion that this could further be explored if more time was available.

The review panel supports the assessment panel's own research recommendations, which include: a) Investigating ways to set up reproductive timing in Stock Synthesis (different versions) and to investigate sensitivities to different choices. This appears to be an important, but largely a technical issue. b) Studying the effect of recreational management actions on the length compositions. c) Investigating different ways to parametrize selectivity. In addition to the suggestions by the assessment panel, which are simpler functions and more informed priors, a suggestion could be to look into formulations based on random effects (state-space models). This allows flexible models for selectivity with few model parameters by setting up processes (e.g. for F at a given length), then the only model parameters to be estimated are only the level and standard deviation of the processes. A model based on this principle (Nielsen and Berg 2014) is routinely used in many ICES assessments and another such model has recently been developed at the Northeast Fisheries Science Center (<https://github.com/timjmiller/wham>). d) Investigating the proportionally few large sharks observed compared to the number of large sharks estimated to be in the population. This apparent dome-shaped selectivity can be caused by a number of different things including spatial distribution. It would be useful to report this "cryptic biomass" to monitor if it is e.g. increasing over time. Further this also relates to flexible modelling of the selectivity (see c above). e) Improved model diagnostics. This and future assessments would benefit, and be simpler to evaluate, if a standard set of model diagnostics were

developed and provided. These could include: residuals (already provided, but should be decorrelated), retrospective analysis, leave-out analysis, jitter-analysis, and simulation validation.

In addition the review panel have suggested a number of research recommendations under TOR 8.

b. Provide recommendations on possible ways to improve the SEDAR process.

The SEDAR process for this meeting was well organized. The meeting was efficient. The assessment panel was able to quickly answer questions and produce new runs and requested diagnostics. So within the constraints imposed by covid this meeting was close to optimal. The support staff was excellent and very helpful.

The presentation team can help the review team by preparing focused presentations, as they are easier to follow (larger fonts, more figures, and less text) than on screen browsing of assessment reports.

Having an assessment review online is not a good substitute for an actual review meeting. The discussion is slower, and hence fewer issues are raised. Also you cannot easily stand up and make an illustrative drawing where needed. Furthermore, the sharing of knowledge, which for other review meetings has been substantial (e.g. sharing tips and tricks of modelling, or introduction to new tools or software) does not happen if all breaks are in isolation. Having informal discussions in person is much better for networking between assessment panel and reviewers, and overall makes the meetings more pleasant and productive.

It might be useful for the chairs of the data workshop working groups to attend parts of the review panel to answer specific questions.

2. TOR 8 Provide suggestions on key improvements in data or modeling approaches that should be considered when scheduling the next assessment.

In addition to the recommendations from the data and assessment panels discussed above, and the recommendations in section 6, the following improvements are recommended for the next assessment.

Recommendations for research activities:

- Species and fleet specific conversions between dressed weight and whole weight should be considered.
- A multi-species analysis of catch rates in the recreational fishery might be useful to extract an abundance index that is not biased by the issues with identification of sharks that are released alive.

- Longevity is poorly estimated and is one basis for estimates of M . Better estimates of longevity, and an independent estimate of natural mortality, for example from a tagging study, would be useful.
- The data workshop discussed whether blacktip sharks may be migrating northward. This migration could be modeled in a spatially explicit assessment. Spatially explicit models might also be useful for explaining differences in trends in indices from different locations.
- The apparent dome shaped selectivity in several gears implies that there are sharks in the population that are older than the oldest individual observed. Whether this is realistic could be validated with fishery independent research.

Recommendations for improvements to data for the assessment:

- The lack of data on catches and size distribution of catches during the peak of the fishery in the 1980s remains a key uncertainty in this assessment. Future work to improve catch reconstruction or evaluate model sensitivity to the catch reconstruction is recommended.
- There is a need to better characterize the length composition, particularly in recreational fisheries, which may be influenced by both state and federal regulations.

Recommendations to the assessment methods:

- Model runs that do not fix parameters should be explored to more accurately characterize the uncertainty in parameter estimates. For example, if there is not enough data to estimate a selectivity parameter for two time blocks, rather than estimating it in one time block and applying the estimated value as a fixed parameter in the other, the data from both time blocks can be pooled to estimate the parameter.
- Bootstrapping the data could be used to quantify the uncertainty contained in the data. Current estimates of uncertainty are conditional on the full dataset and the modeling assumptions.
- Projections should be done using MCMC or profile likelihood methods to evaluate whether the normal approximation was adequate. [assessment]
- Further research is needed on inconsistency of indices including the hierarchical models considered in this analysis.
- Improved model diagnostics are needed, as recommended in the assessment panel report and as described under TOR 6a.
- Explore whether some other functional form of the stock recruit relationship would be more appropriate for this species, such as the low fecundity model that was used in the low productivity sensitivity. Explore using reference points that don't depend on MSY such as SPR -based reference points.
- Investigate the timing and duration of the recruitment period, the duration of age 0 natural mortality, and the possibility of age 0 catches occurring during the recruitment period.